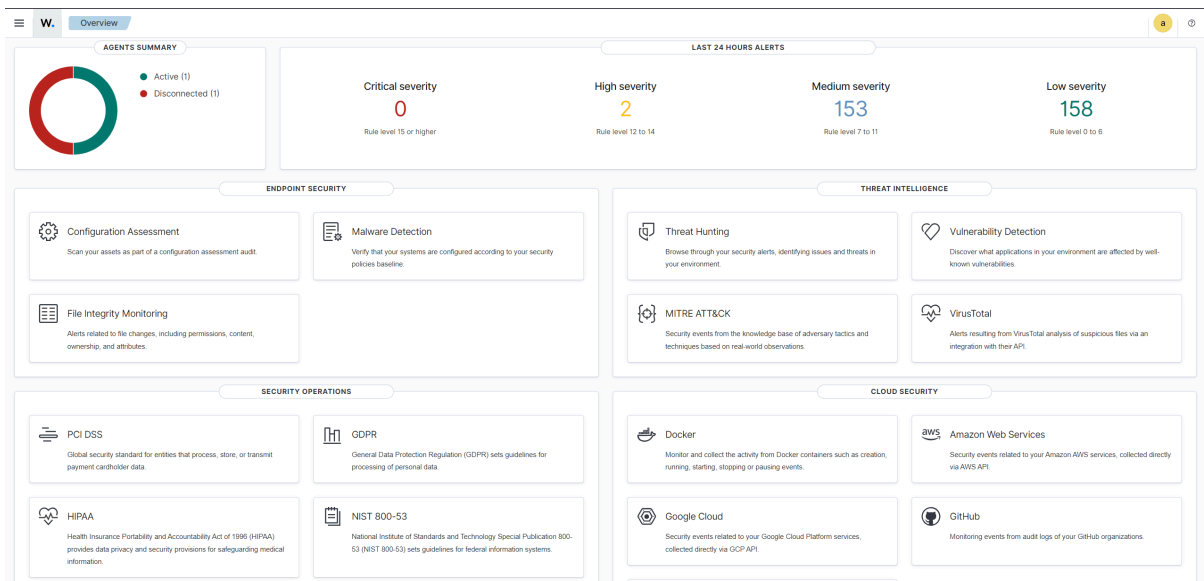


Attack Scenarios and Detection Engineering

Wazuh Dashboard Deployed, using Wazuh agent (CT1-node)



Scenario 1: Authentication Brute-Force Attack

- **MITRE ATT&CK: T1110.001 - Credential Access: Password Guessing / Brute Force**

1. Attack Execution (Red Team):

From an external host on the network, rapid automated SSH login attempts were executed against non-existent and existing accounts using bad credentials:

codeBash

```
# Simulating rapid failed authentications
for i in {1..6}; do ssh fakeuser@<TARGET_IP> -o BatchMode=yes; done
```

2. Detection & Telemetry (Blue Team):

- **Log Source:** /var/log/auth.log
- **Triggered Rules:**
 - **Rule 5710:** sshd: Attempt to login using a non-existent user (Severity 5)
 - **Rule 5712:** sshd: brute force trying to get access to the system (Severity 10)
- **SOC Triage:** The Wazuh Manager aggregated the sequential authentication failures originating from a single source IP, escalated the severity based on frequency thresholds, and generated a High-Severity security alert.

rule.description	sshd: brute force trying to get access to the system. Non existent user.
rule.firedtimes	1
rule.frequency	8
rule.gdpr	IV_35.7.d, IV_32.2
rule.groups	syslog, sshd, authentication_failures
rule.hipaa	164.312.b
rule.id	5712
rule.level	10
rule.mail	false
rule.mitre.id	T1110
rule.mitre.tactic	Credential Access
rule.mitre.technique	Brute Force
rule.nist_800_53	SI.4, AU.14, AC.7
rule.pci_dss	11.4, 10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	Sep 7, 2026 @ 18:15:03.196

Scenario 2: Unauthorized File Tampering & Persistence (FIM)

- **MITRE ATT&CK: T1053.003 - Persistence: Scheduled Task/Job: Cron**

1. Attack Execution (Red Team):

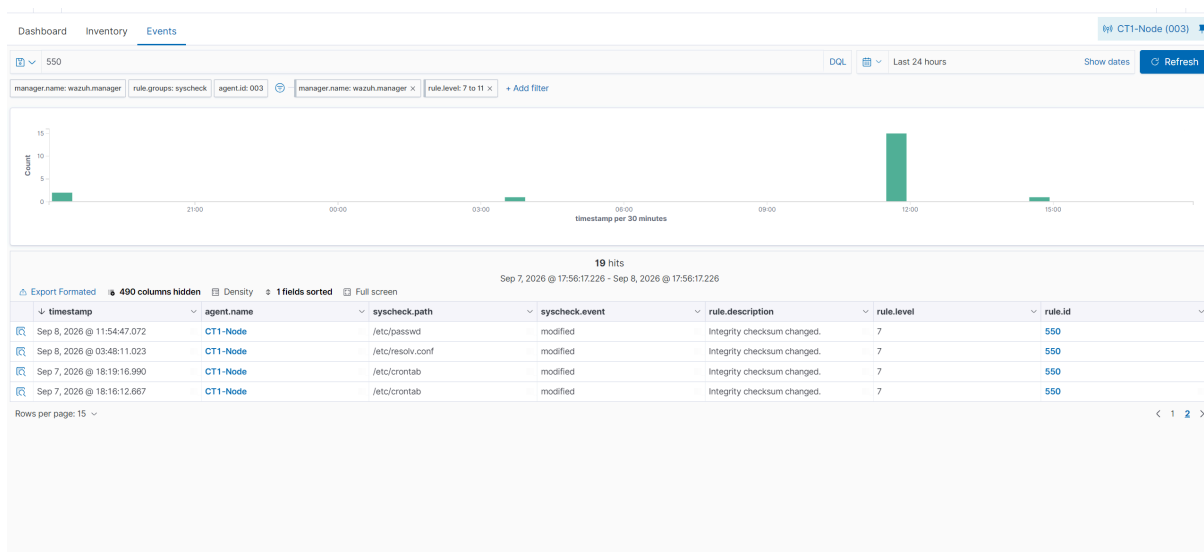
Simulating an adversary who gained initial access and attempts to establish persistent backdoor execution via system crontabs:

codeBash

```
# Injecting a malicious backdoor persistence task
echo "* * * * * root /tmp/evil_backdoor.sh" >> /etc/crontab
```

2. Detection & Telemetry (Blue Team):

- **Detection Mechanism: Wazuh Syscheck (File Integrity Monitoring)** configured with realtime="yes" for /etc.
- **Triggered Rule: Rule 550:** Integrity checksum changed (Severity 7)
- **Forensic Evidence Captured:**
 - **Target File:** /etc/crontab
 - **Action:** Modified
 - **Hash Comparison:** Real-time SHA256 checksum mismatch recorded before and after injection.
 - **File Diff:** Exact line modification identified: + * * * * * root /tmp/evil_backdoor.sh.



Scenario 3: Privilege Escalation via Sudo (Custom Detection Rule)

- **MITRE ATT&CK: T1548.003 - Privilege Escalation: Abuse Elevation Control Mechanism: Sudo**

1. Threat Context:

Adversaries often attempt to escalate privileges from a compromised low-privilege service or user account by executing interactive shells using sudo su or sudo -i.

2. Attack Execution (Red Team):

A standard non-root user (john) was utilized to execute an interactive root elevation:

codeBash

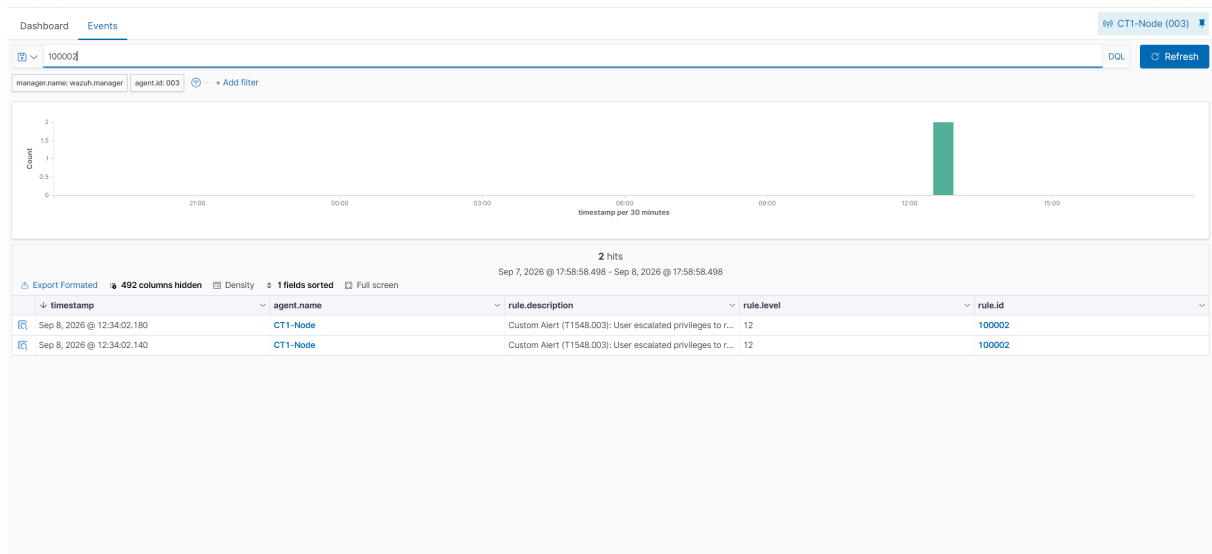
```
su - john  
sudo su
```

3. Custom Rule Engineering (Detection Engineering):

A custom rule was created in /var/ossec/etc/rules/local_rules.xml to flag interactive root shell spawning:

codeXml

```
<group name="custom_priv_escalation,">  
  <rule id="100002" level="12">  
    <if_group>sudo</if_group>  
    <match>su</match>  
    <description>Custom Alert (T1548.003): User escalated p  
rivileges to root via sudo su</description>  
    <mitre>  
      <id>T1548.003</id>  
    </mitre>  
  </rule>  
</group>
```



t	agent.name	CT1-Node
t	decoder.ftscoment	First time user executed the sudo command
t	decoder.name	sudo
t	decoder.parent	sudo
t	full_log	Sep 08 02:34:01 CT1 sudo[14103]: pam_limits(sudo:session): Could not set limit for 'core' to soft=0, hard=-1: Operati on not permitted; uid=0,euid=0
t	id	1788834842.54698
t	input.type	log
t	location	journald
t	manager.name	wazuh.manager
t	predecoder.hostname	CT1
t	predecoder.program_name	sudo
t	predecoder.timestamp	Sep 08 02:34:01
t	rule.description	Custom Alert (T1548.003): User escalated privileges to root via sudo su
#	rule.firedtimes	2
t	rule.groups	custom_priv_escalation
t	rule.id	100002
#	rule.level	12
🔍	rule.mail	true
t	rule.mitre.id	T1548.003
t	rule.mitre.tactic	Privilege Escalation, Defense Evasion
t	rule.mitre.technique	Sudo and Sudo Caching
📅	timestamp	Sep 8, 2026 @ 12:34:02.180